

Input

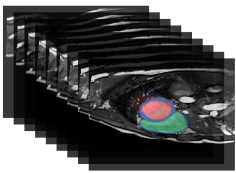
Initialization

Deformation

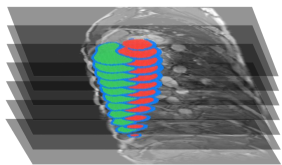
Occupancy Computation

Supervision

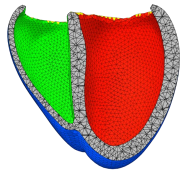
MRI Slices



Slice Labels

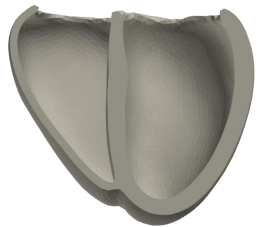


Initial Mesh



1

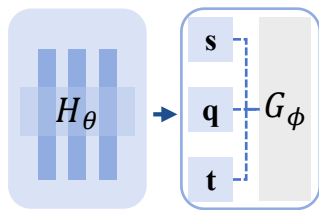
Initial



$$M_0 = (P_0, C, R)$$

2

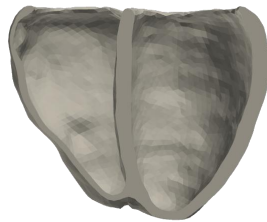
Real-NVP + Affine



$$G_\phi(H_\theta)$$

3

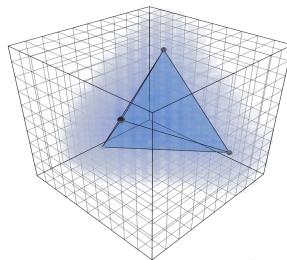
Deformed Mesh



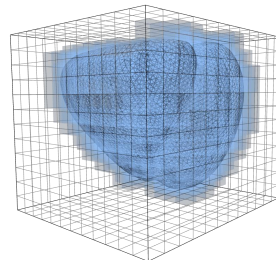
$$M_{\theta, \phi} = (P_{\theta, \phi}, C, R)$$

4

Tet Occupancy

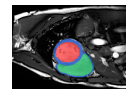


$$x_k = \prod_{j=1}^4 \sigma(\alpha(h_{k,j}(x) + \beta))$$

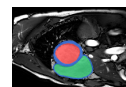
Block-wise
Occupancy

Slice Supervision

GT

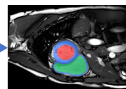


⋮

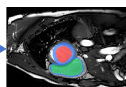


⋮

Prediction



⋮



⋮

Occ Loss



⋮

Occ Loss



⋮

$$\mathcal{L}_{occ} = \mathcal{L}_1 + \mathcal{L}_{dice}$$



Loss

$$\mathcal{L}_{lap} + \mathcal{L}_{qual} + \mathcal{L}_{occ}$$

Backpropagation to θ, ϕ